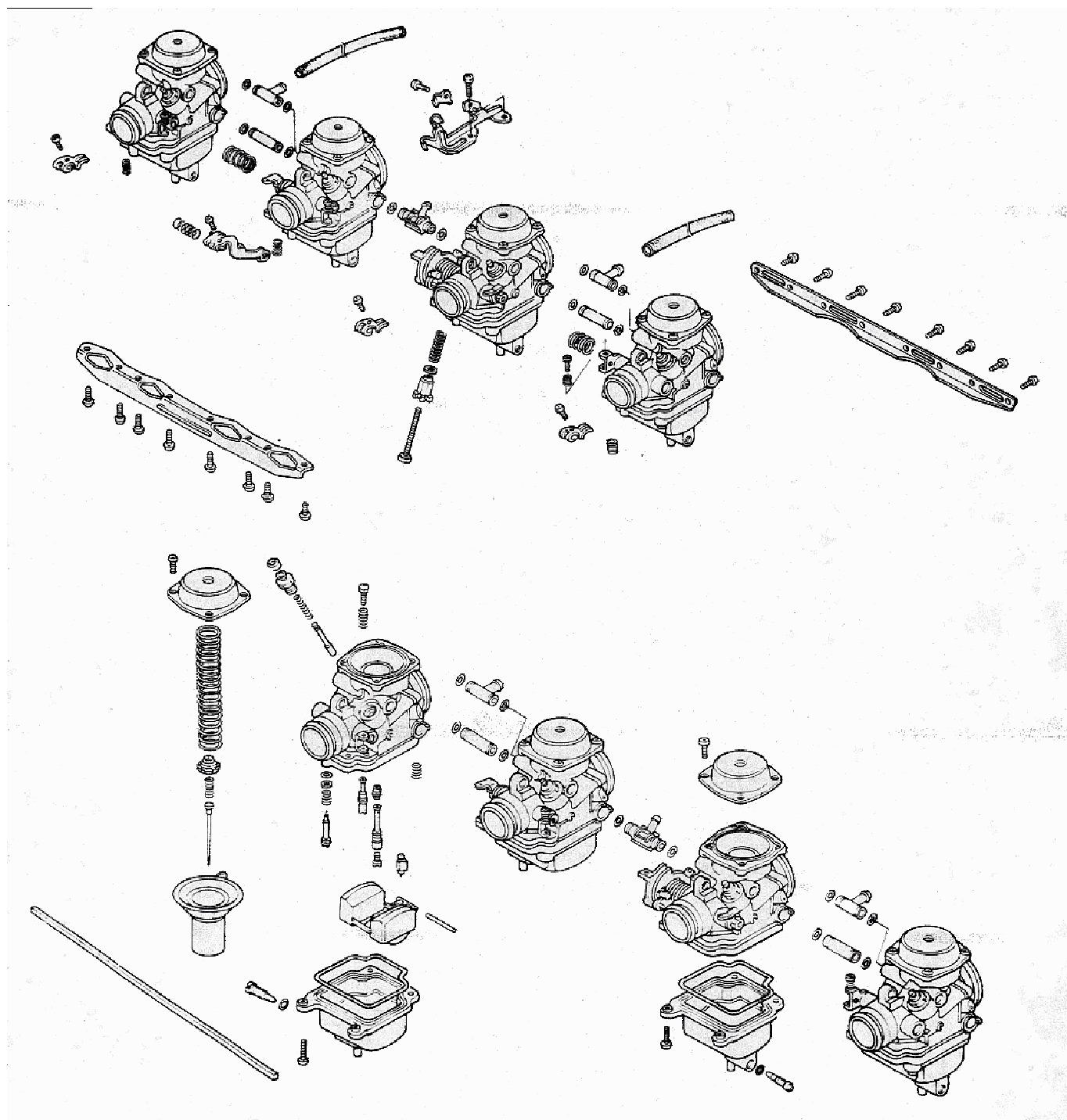




HONDA CBX750F

4. Fuel System





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Service Information

General

WARNING

Gasoline is extremely flammable and is explosive under certain conditions. Work in well-ventilated area. Do not smoke or allow flames or sparks in the work area.

- When disassembling fuel system parts. Note the location of the O-rings. Replace them with new ones on reassembly.
- The float bowls have drain screws that can be loosened to drain residual gasoline.

SPECIFICATIONS

Item	Specifications
Carburettor Type	KEIHIN VE
Throttle Bore	34 mm (1.34 in)
Venturi Bore	30.8 mm (1.21 in)
Identification No.	VE64B
Float Level (Gauge Level)	18.5 mm (0.73 in)
Main Jet	#112
Idle Speed	1,000 +/- 100 rpm
Throttle grip free play	2-6 (0.008-0.24 in)
Pilot Screw Initial Opening	2 Turns Out

TOOLS

Special

Carburettor pilot screw wrench 07908 - 4220201

Common

Float gauge 07401 - 0010000



TROUBLESHOOTING

Engine cranks but wont start

1. No fuel in tank
2. No fuel in carburettor
3. Engine flooded with fuel
4. No spark at plug (ignition malfunction)
5. Air cleaner clogged
6. Intake air leaking
7. Improper choke operation
8. Improper throttle operation

Hard starting or stalling after starting

1. Improper choke operation
2. Ignition malfunction
3. Fast idle speed incorrect
4. Carburettor malfunction
5. Fuel contaminated
6. Intake air leak
7. Idle speed incorrect

Rough Idle

1. Ignition malfunction
2. Idle speed incorrect
3. Incorrect carburettor synchronisation
4. Carburettor malfunction
5. Fuel contaminated

Misfiring during acceleration

Ignition malfunction

Backfiring

1. Ignition malfunction
2. Carburettor malfunction

Poor performance (Driveability) and poor fuel economy

1. Fuel system clogged
2. Ignition malfunction

Lean Mixture

1. Clogged fuel jets
2. Piston stuck closed
3. Faulty float valve
4. Float level low
5. Fuel cap vent blocked
6. Fuel strainer screen clogged
7. Restricted fuel line
8. Air vent tube clogged
9. Intake air leak

Rich mixture

1. Clogged air jets
2. Faulty float valve
3. Float valve too high
4. Choke stuck closed
5. Dirty air cleaner

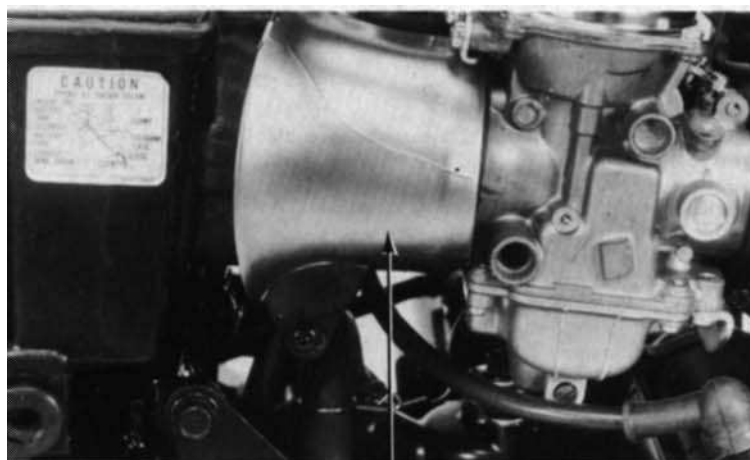


CARBURETOR REMOVAL

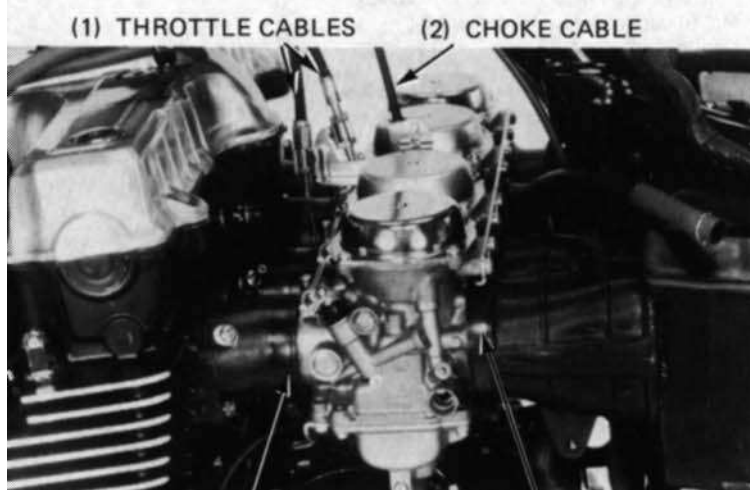
Remove fuel tank (page 4-15)
Remove the left and right connecting tube covers.

Loosen the air cleaners connecting tube bands and carburettor intake pipe bands. Remove the choke and throttle cables from the bracket.

Remove the battery holder.



(1) CONNECTING TUBE COVER

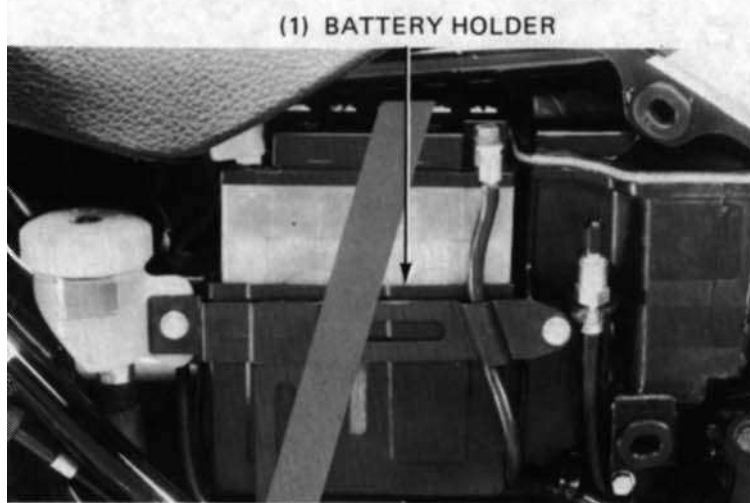


(1) THROTTLE CABLES

(2) CHOKE CABLE

(3) INTAKE PIPE BAND

(4) CONNECTING TUBE BAND



(1) BATTERY HOLDER

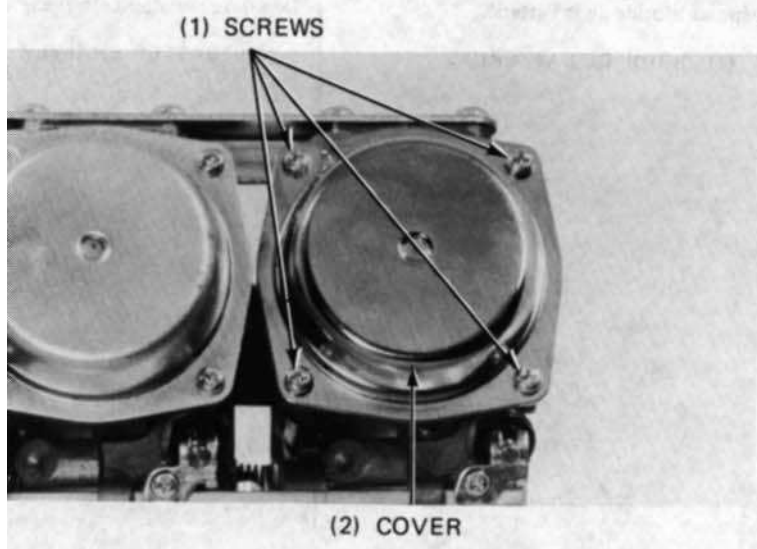
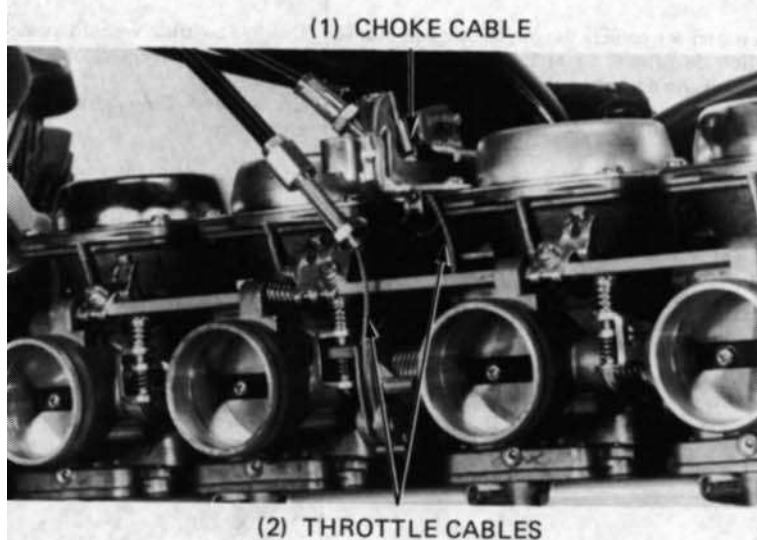
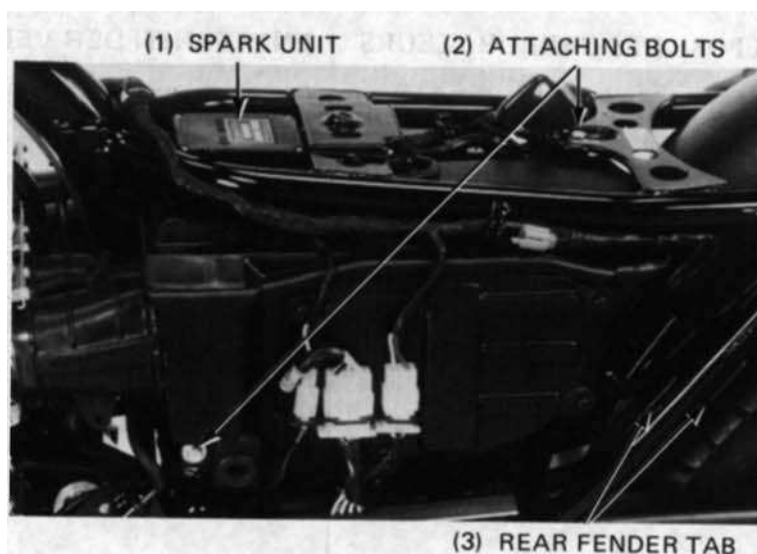


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4. Fuel System

Remove the bolts attaching the air cleaner case to the frame.
Remove the spark unit from the air cleaner case.
Unhook the rear fender tabs from the frame.
Slide the air cleaner rearward.

Remove the carburettor assembly from the left side and disconnect the choke and throttle cables from the carburettor.



VACUUM CHAMBER

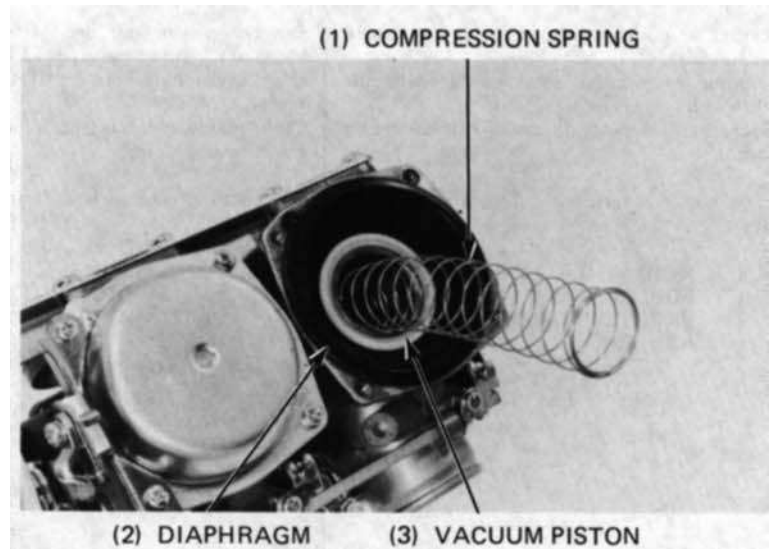
Removal

Remove the four vacuum chamber cover screws and cover.

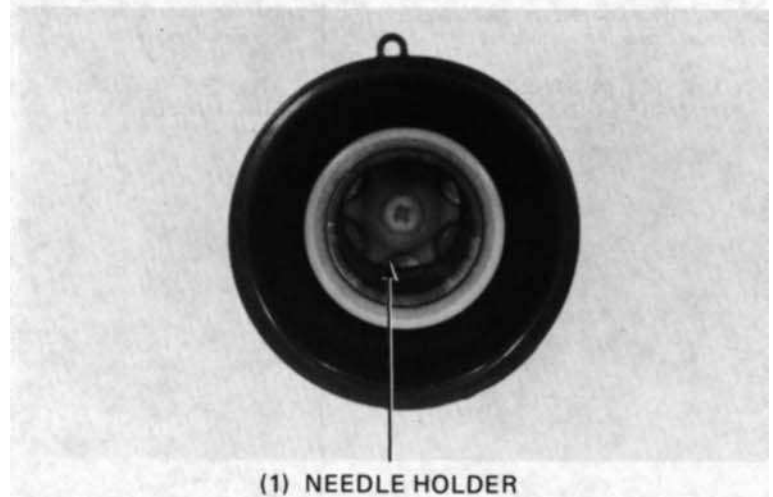


4. Fuel System

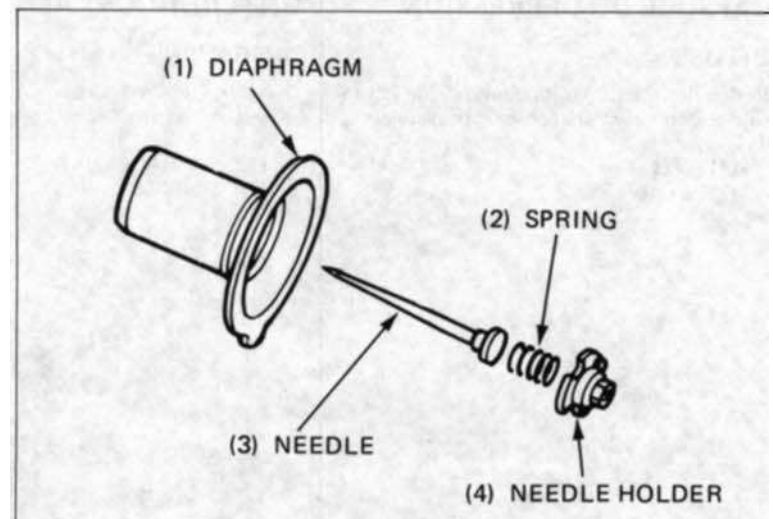
Remove the compression spring, diaphragm and vacuum piston. Inspect the vacuum piston for wear, nicks, scratches or other damage. Make sure the piston moves up and down freely in the chamber.



Push the needle holder in and turn it 60 degrees with a 8 mm socket. Then remove the needle holder, spring and needle from the piston.



Inspect the needle for excessive wear at the tip and for bending, or other damage. Check for a torn diaphragm or other deterioration.





Installation

Installation is essentially the reverse of the removal but to keep from distorting the diaphragm, install the vacuum piston/diaphragm as follows:

Insert the vacuum piston into the carburettor. Stick your finger into the carburettor bore and hold the vacuum piston in the full throttle position, then turn down the diaphragm so its lip fits into the carburettor groove.

Install the chamber cover, aligning its cavity with the hole in the carburettor, and secure with at least two screws before releasing the vacuum piston.

FLOAT CHAMBER

Removal

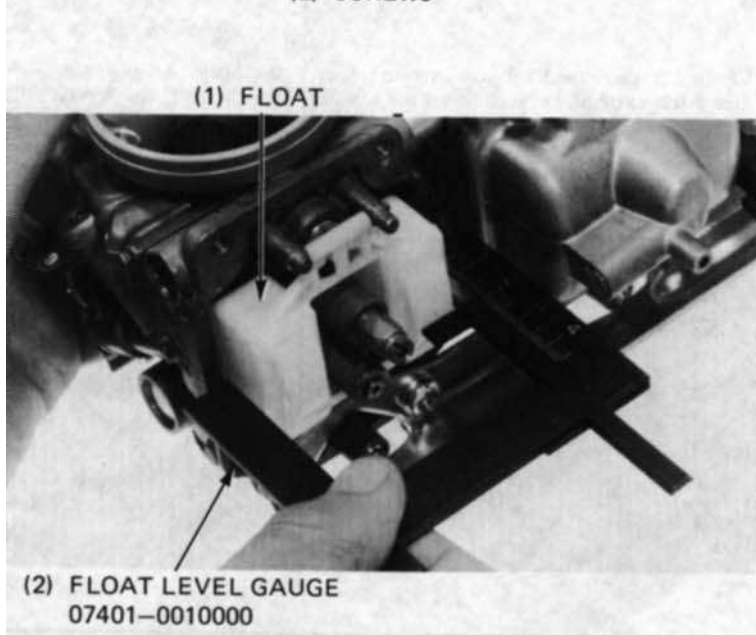
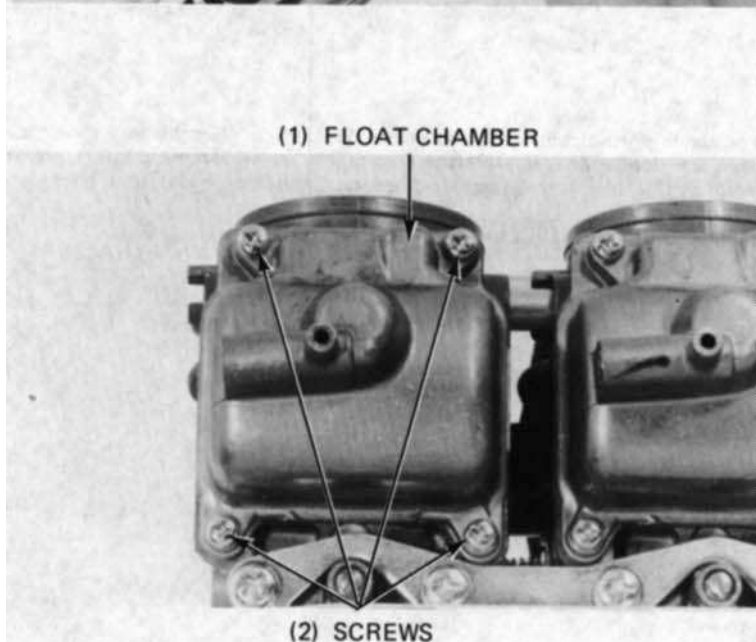
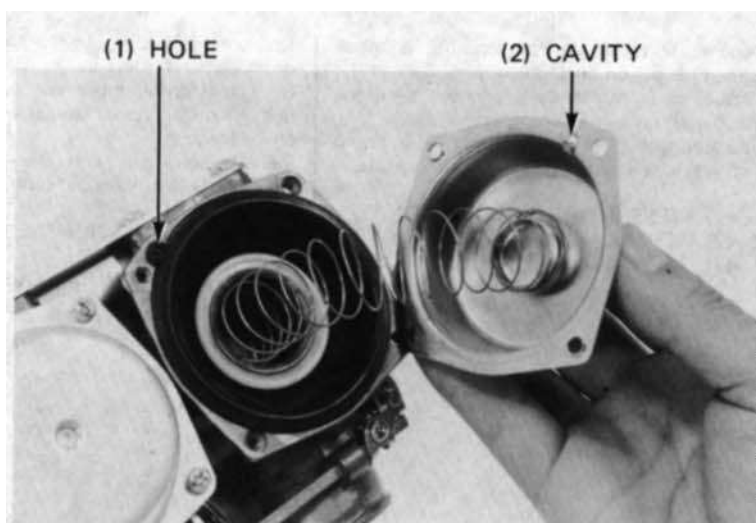
Remove the four float chamber screws and the float chamber.

FLOAT LEVEL

Measure the float level with the float tang just contacting the float valve.

Float Level: $18.5 \pm 1\text{mm}$ (0.61 ± 0.04 in)

Replace the float assembly, if it is not within specifications.





4. Fuel System

Float and Jets

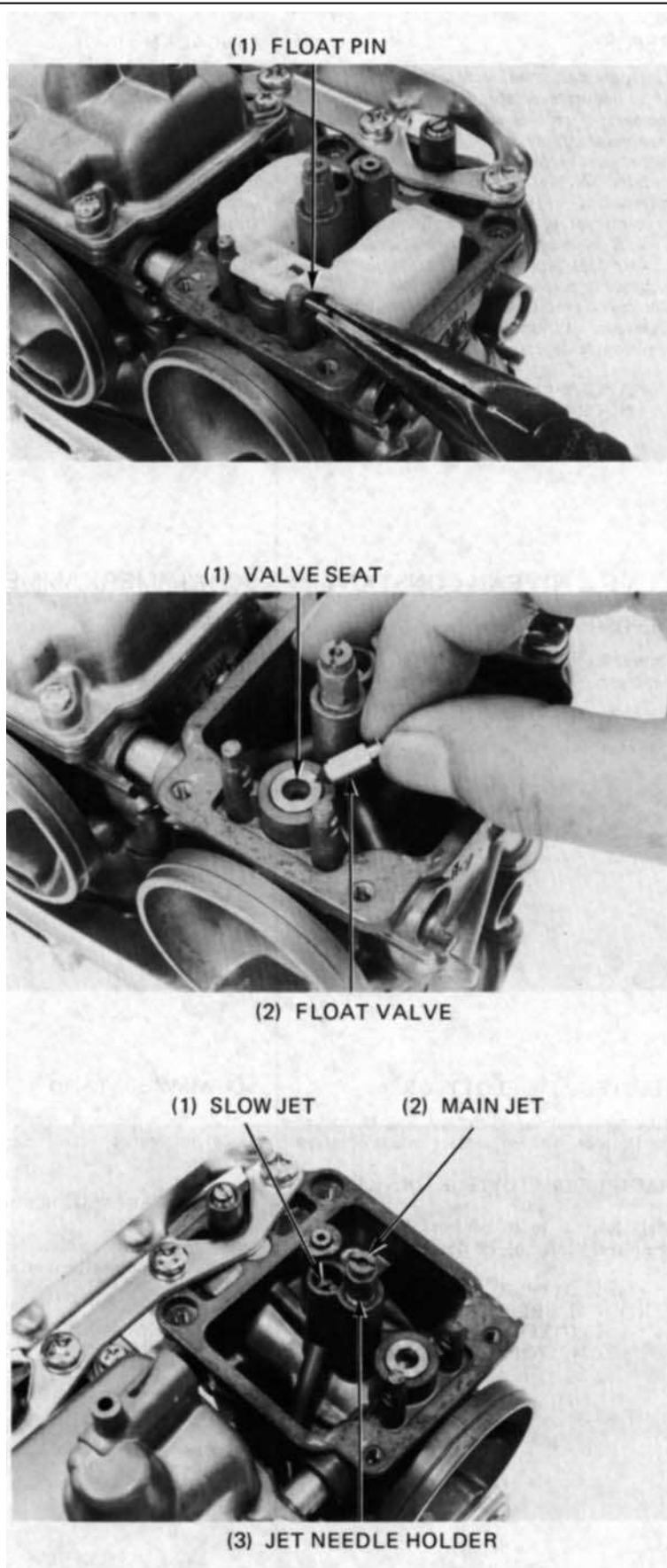
Remove the float pin, float and float valve.

Inspect the float valve for grooves and nicks.
Inspect the operation of the float valve.

Remove the main jet, slow jet and the jet needle holder.

ASSEMBLY

Assemble the float chamber components in the reverse order of disassembly.





4. Fuel System

CHOKE (BYSTARTER) VALVE

Removal

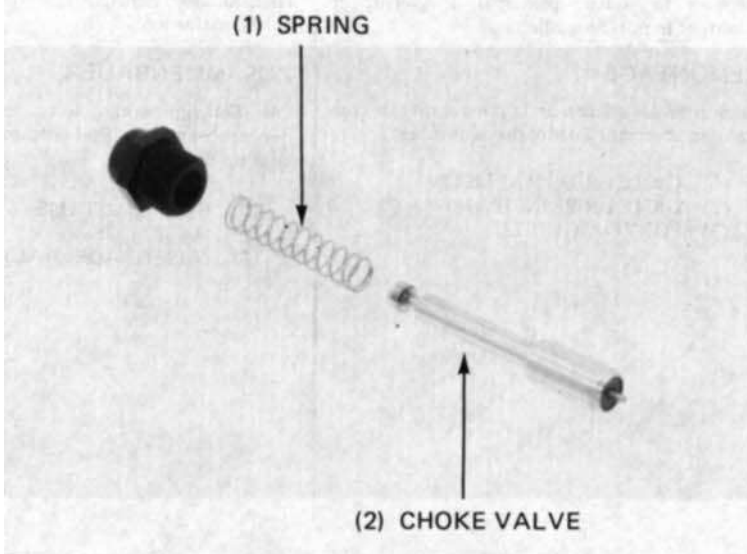
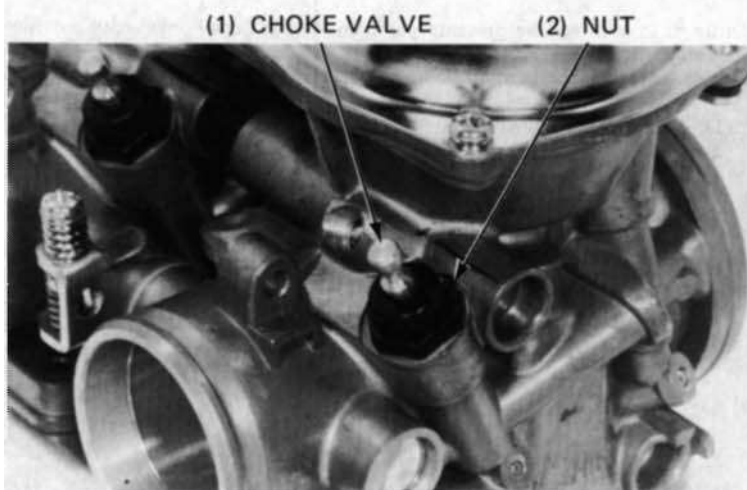
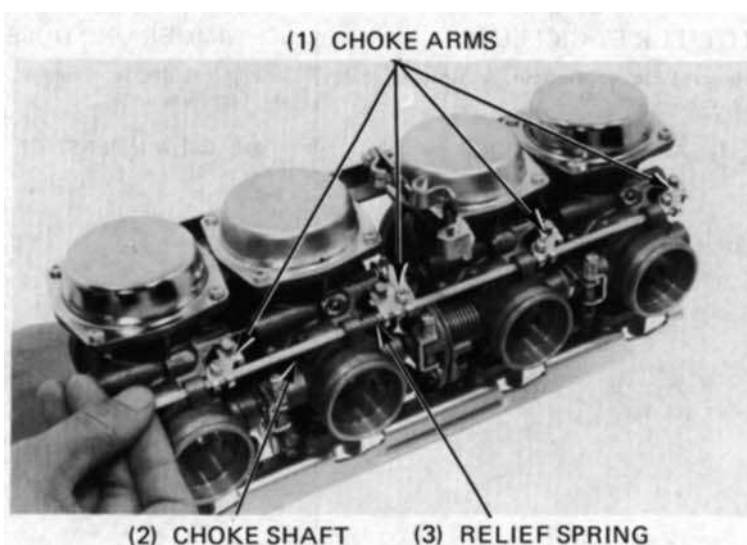
Unhook the choke (bystarter) valve relief spring from the No. 3 carburettor choke arm.

Loosen all choke (bystarter) arm clamp screws.

Pull the choke shaft out from the right side and remove the choke arms and relief spring.

Remove the choke valve nut, spring and valve.

Check the choke valve and spring of nicks, grooves or other damage.





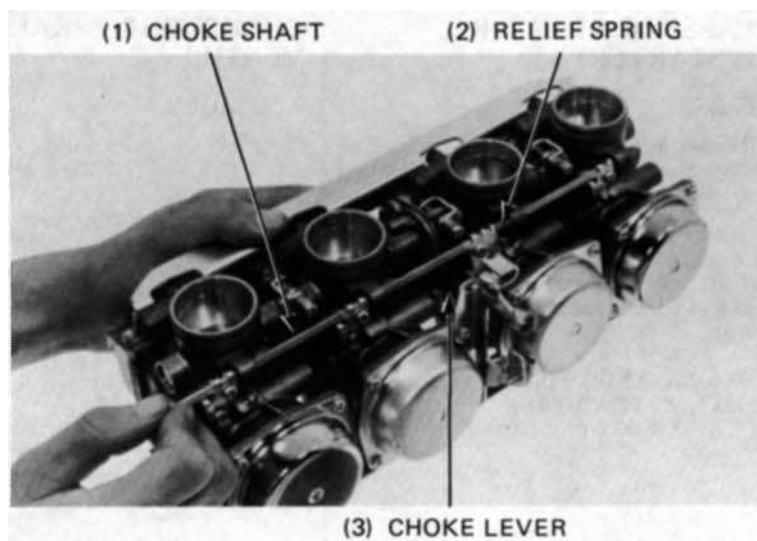
HONDA CBX750F

4. Fuel System

Installation

Install the choke valve, spring and nut. Install the choke arm and relief spring, and insert the choke shaft from the left side.

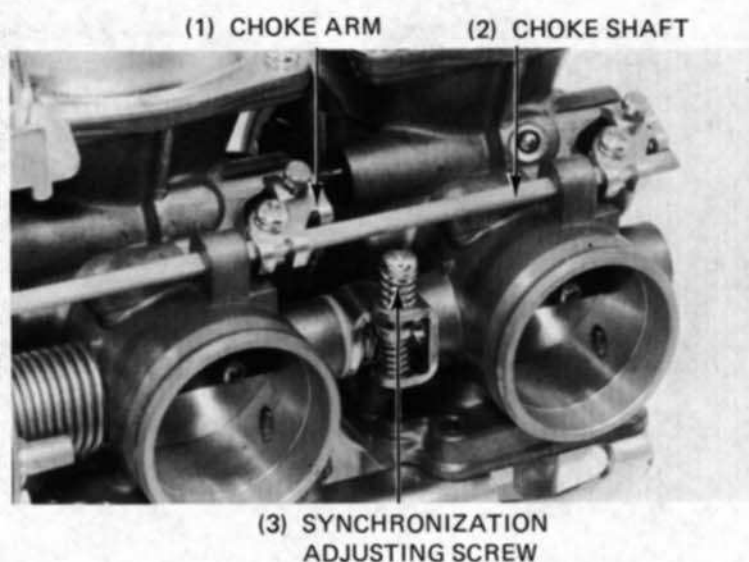
Tighten the choke arm clamp screws. Hook the choke relief spring correctly. Make sure that the choke linkage operates smoothly by moving the choke lever.



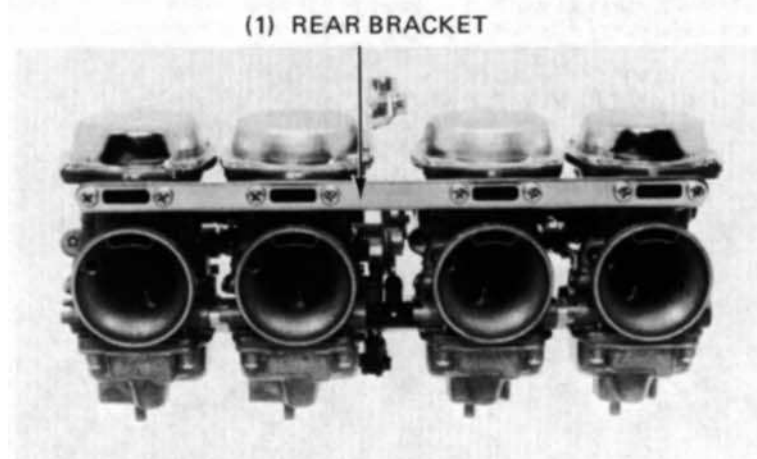
CARBURETOR SEPARATION

Remove the choke shaft, arms and relief spring (page 4-8).

Loosen each carburettor's synchronisation adjusting screw until there is no spring tension on it.



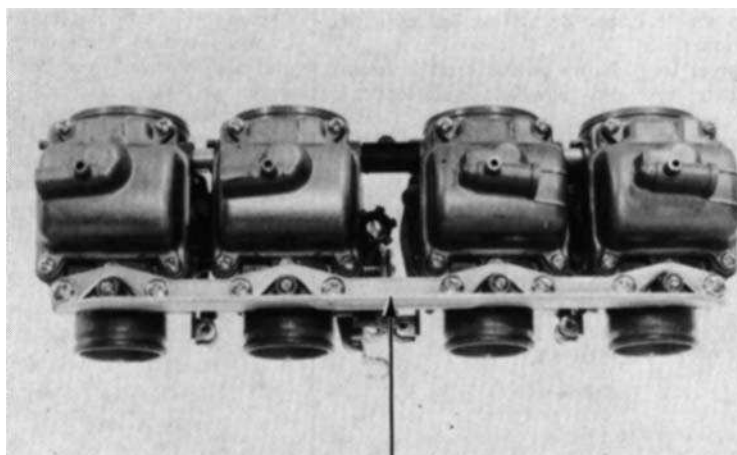
Remove the rear bracket.





4. Fuel System

Remove the front bracket.

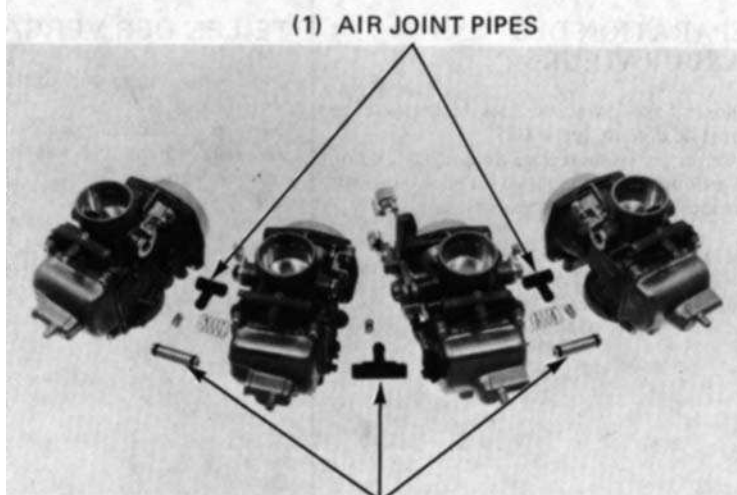


(1) FRONT BRACKET

Carefully separate the carburetors.

CAUTION

Separate the carburetors horizontally to prevent damage to the fuel and air joints pipes.



(1) AIR JOINT PIPES

(2) FUEL JOINT PIPES

CARBURETTOR CLEANING

Remove the vacuum piston (page 4-4).

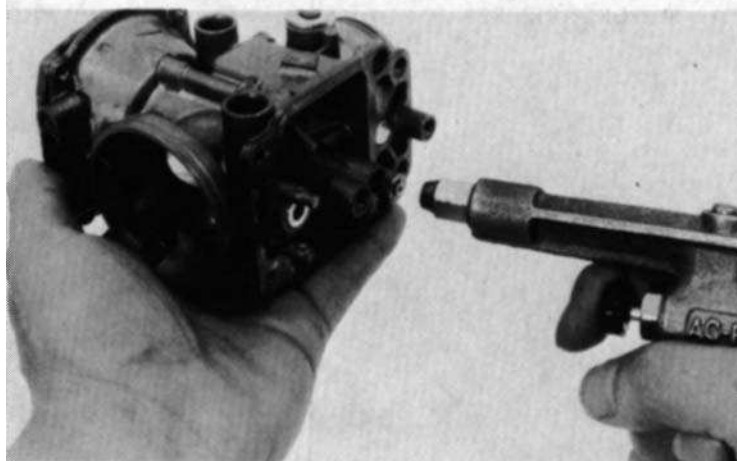
Remove the float, float valve and jets (page 4-7).

Remove the choke valve (page 4-8).

Remove the pilot screw (page 4-13).

Clean the carburettor passages with compressed air.

Install the removed parts.



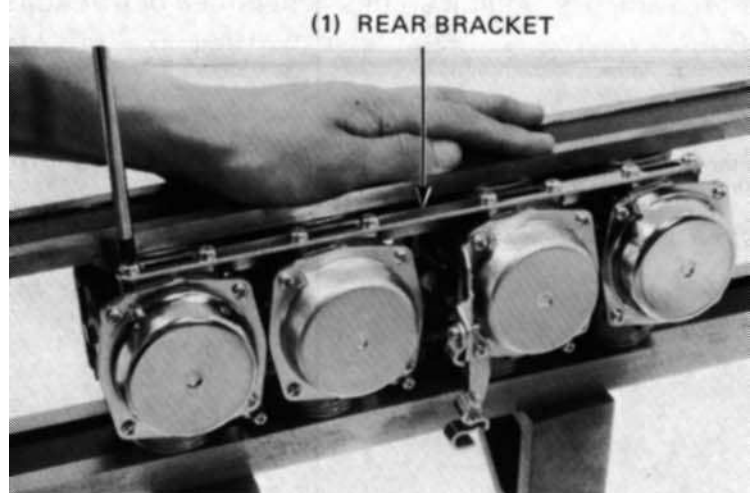
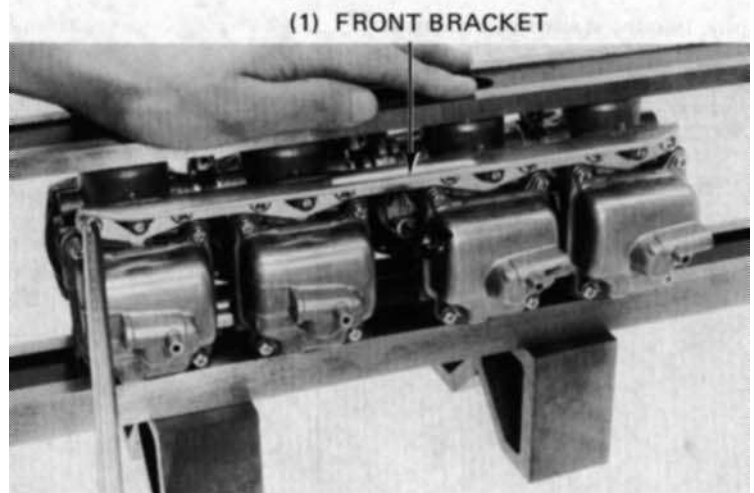
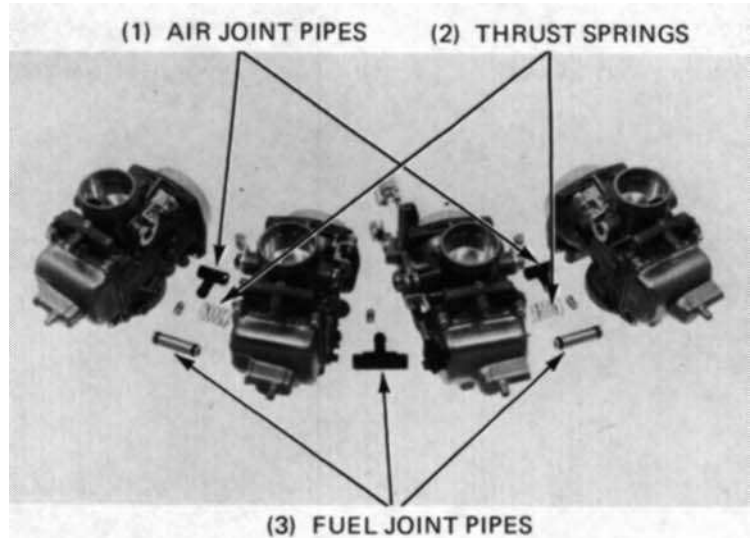


CARBURETTOR ASSEMBLY

Coat new O-rings with oil and install them on the fuel and air joint pipes.
Connect the carburetors with the fuel and air joint pipes.
Install the thrust springs between the throttle valve shafts.

Install the front bracket loosely.
Place the carburettor on a flat surface with the front side facing up.
Press the carburetors together carefully and tighten the front bracket screws in the sequence shown in two or three steps to prevent carburettor misalignment.

Install the rear bracket in the same procedure as for the front bracket.





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4. Fuel System

Install the choke arms and relief spring. Carefully insert the choke shaft through the choke arms and relief spring from the left-hand side.

Tighten the choke arm clamp screws and hook the relief spring correctly.

Install the synchronisation springs. Turn the throttle stop screw to align the No. 2 throttle valve with the edge of the by-pass hole.

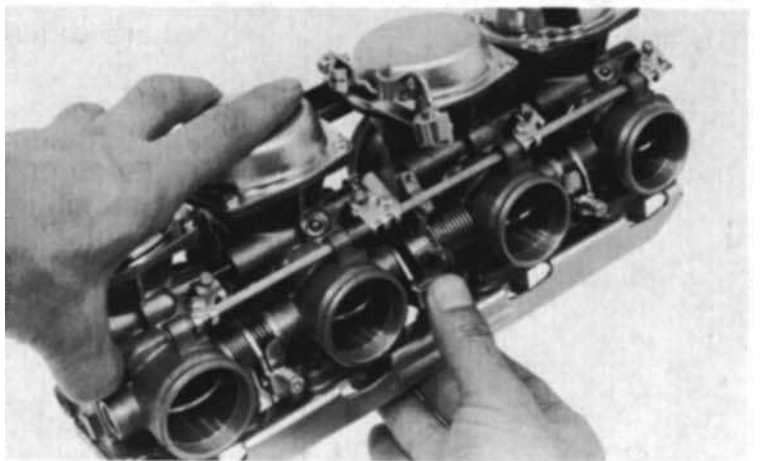
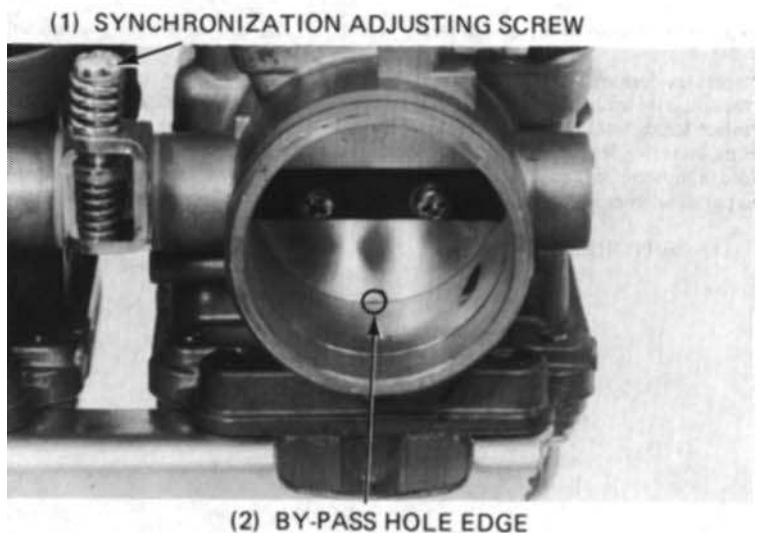
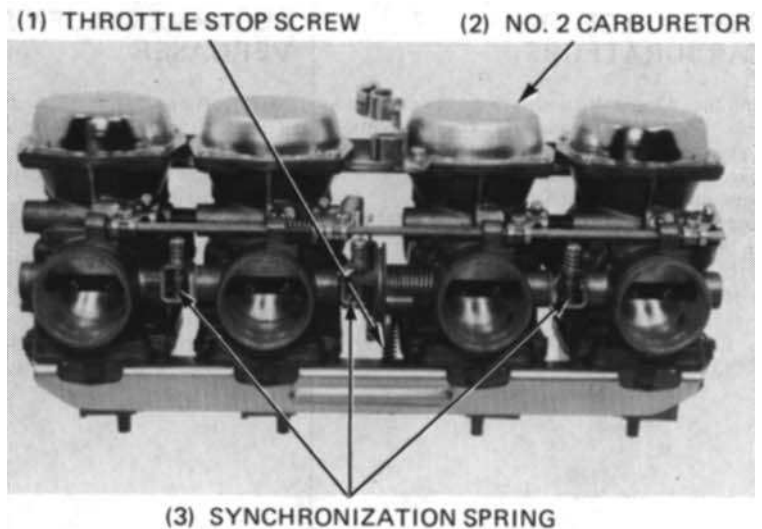
Align each throttle valve with the edge of the bypass hole by turning the synchronisation adjusting screw.

Inspect throttle operation as described below:

- Open the throttle slightly by pressing the throttle linkage. Then release the throttle.
- Make sure that it returns smoothly.
- Make sure that there is no drag when opening and closing the throttle.

Make sure that the choke valve operation is smooth by moving the choke linkage.

Close the choke valve by turning the choke linkage. Release the choke linkage and make sure that it returns smoothly.





PILOT SCREW

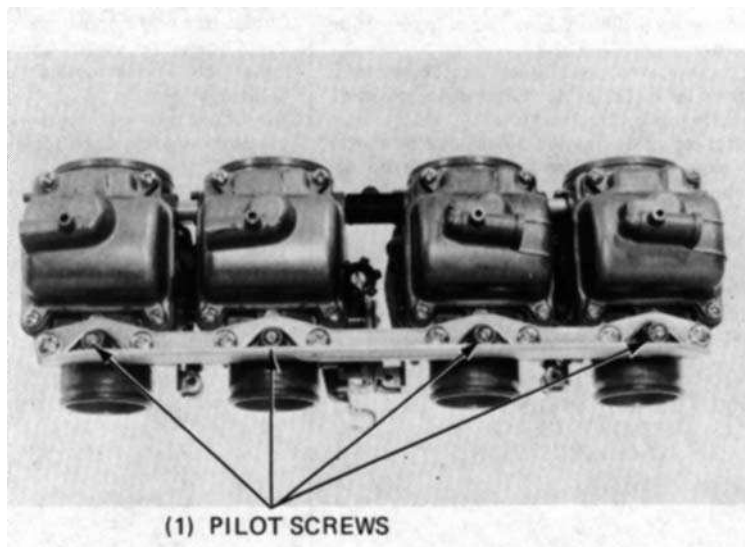
Removal

NOTE:

The pilot screws are factory pre-set and should not be removed unless the carburetors are overhauled.

Turn each pilot screw in and carefully count the number of turns before it seats lightly.

Make note of this to use as a reference when reinstalling the pilot screws.



CAUTION

Damage to the pilot screw seat will occur if the pilot screw is tightened against the seat

Remove the pilot screw and inspect them.
Replace they are worn or damaged.

Installation

Install the pilot screws and return them to their original position as noted during removal. Perform pilot screw adjustment if a new pilot screw is installed (page 4-14).

NOTE:

If you have replaced the pilot screw in one carburettor, you must replace the pilot screws in the other carburetors for proper pilot screw adjustment.

CARBURETOR INSTALLATION

Install is essentially the reverse of removal.

NOTE:

Route the throttle and choke cables properly (page 1-8).

Perform the following inspection and adjustment.

- Throttle operation (page 3-5).
- Carburetor choke (page 3-6)
- Carburetor idle speed (page 3-9)
- Carburetor synchronisation (page 3-8)



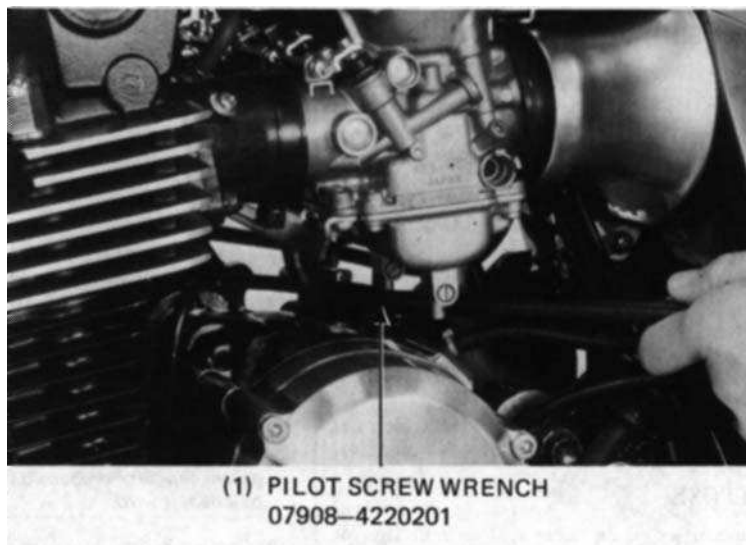
PILOT SCREW ADJUSTMENT

Idle Drop Procedure

NOTE:

- The pilot screws are factory pre-set and no adjustment is necessary unless the pilot screws are replaced (page 4-13).
- Use a tachometer with graduation of 50 min^{-1} (rpm) change.

1. Turn each pilot screw clockwise until it seats lightly and back it out to the specification given. This is the initial setting prior to the final pilot screw adjustment.



INITIAL OPENING: 2 Turns out

CAUTION

Damage to the pilot screw seat will occur if the pilot screw is tightened against the seat.

2. Warm up the engine to operating temperature. Stop and go driving for 10 minutes is sufficient.
3. Attach a tachometer according to the manufactures instructions.
4. Adjust the idle speed with the throttle stop screw.
IDLE SPEED: $1,000 \pm 100 \text{ min}^{-1}$ (rpm)
5. Turn each pilot screw $1/2$ turn out from the initial setting.
6. If engine speed increases by 50 min^{-1} (rpm) or more, turn each pilot screw out by a continual $1/2$ turn until the engine speed drops by 50 min^{-1} (rpm) or less.
7. Adjust the idle speed with the throttle stop screw.
8. Turn the No. 1 carburettor pilot screw in until the engine speed drops 50 min^{-1} (rpm).
9. Turn the No. 1 carburettor pilot screw 1 turn out from the position obtained in step 8.
10. Adjust the idle speed with the throttle stop screw.
11. Perform steps 8, 9 and 10 for the No. 2, 3 and 4 carburettor pilot screws.



FUEL TANK

WARNING

Do not allow flames or sparks near gasoline.

Wipe up spilled gasoline at once.

REMOVAL

Unlock the helmet holder, release the seat by pushing the latch lever forward and remove the seat.

Remove the left and right frame side covers.

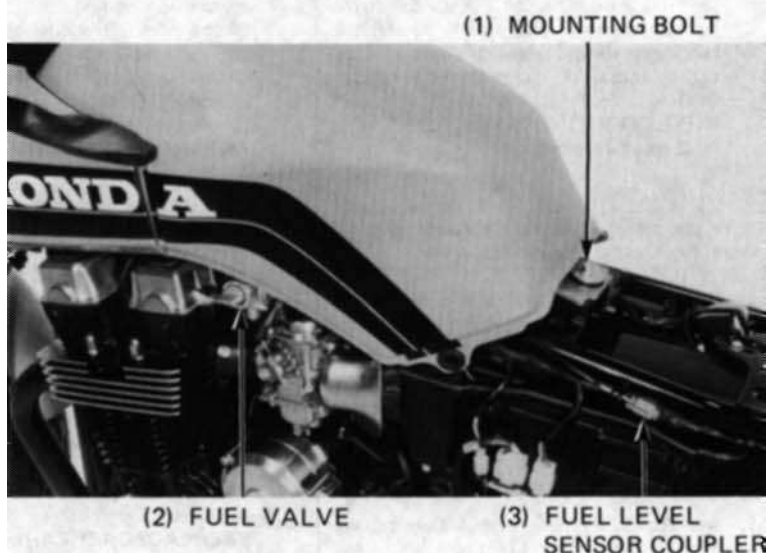
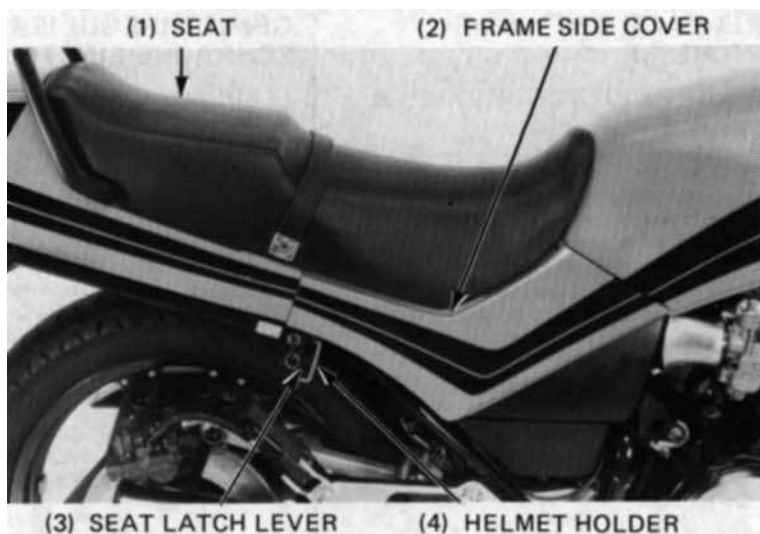
Turn the fuel valve off.

Disconnect the fuel level sensor coupler.

Remove the fuel tank mounting bolt.

Raise the fuel tank and disconnect the fuel valve vacuum hose and fuel hose.

Remove fuel tank from the frame.

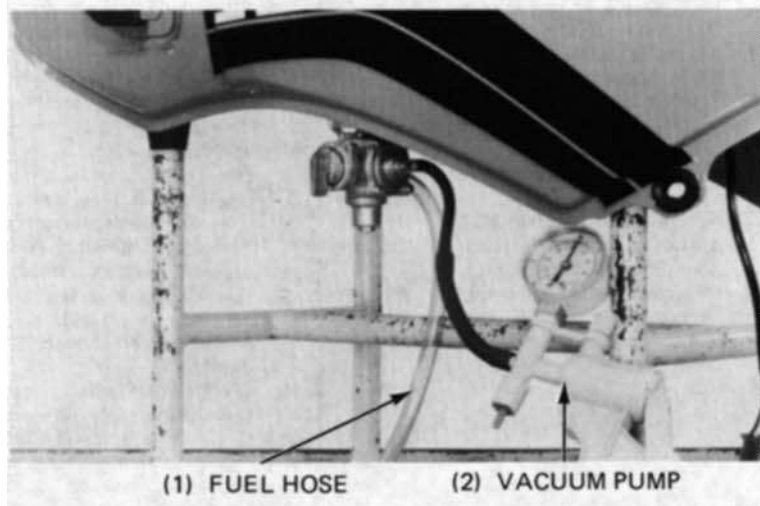


FUEL VALVE DIAPHRAGM TEST

Connect the fuel hose to the fuel valve and place a container under the fuel hose.

Turn the fuel hose on. If fuel comes out of the fuel hose, replace the diaphragm.

Connect a hand vacuum pump to the diaphragm vacuum outlet. Fuel should flow from the fuel hose when vacuum is applied. If fuel flow is restricted, replace the diaphragm.





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4. Fuel System

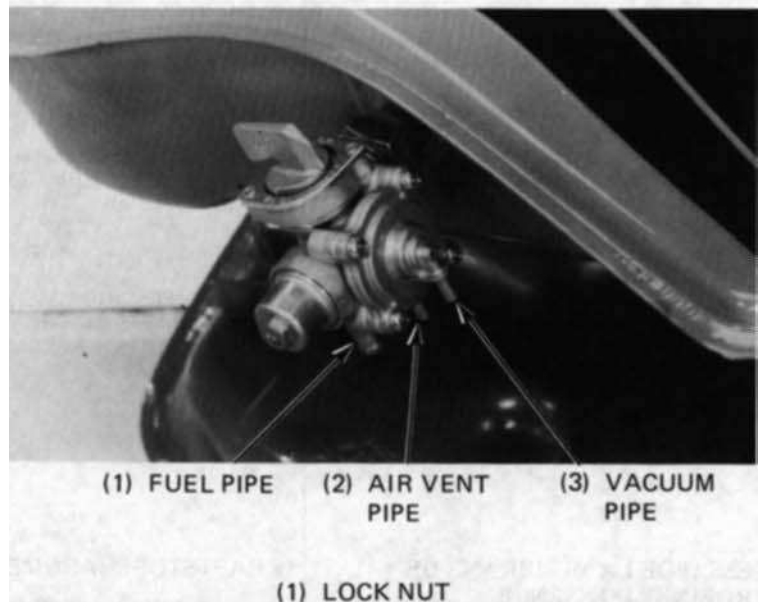
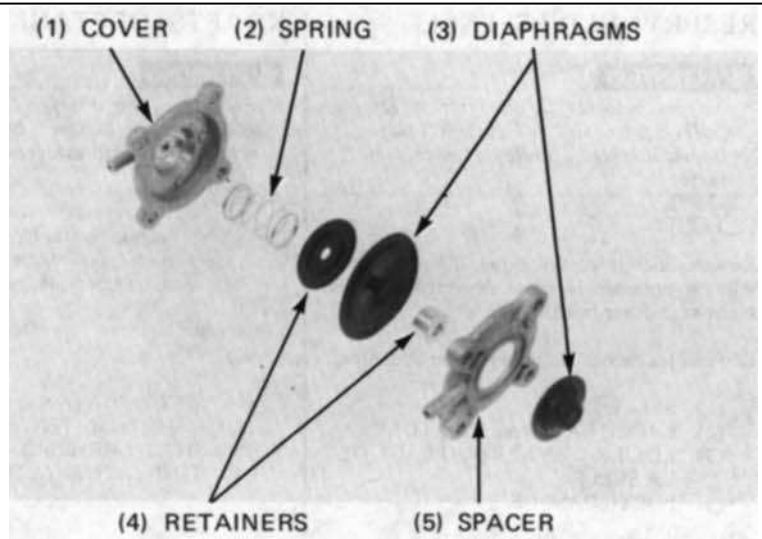
FUEL VALVE DIAPHRAGM REPLACEMENT

Drain the fuel from the fuel tank.
Remove the four attaching screws and the diaphragm assembly.
Replace the diaphragm cover, spring, diaphragms spacer and retainers as a set.

Install the diaphragm assembly so that the air vent pipe of the spacer and the vacuum pipe of the cover face to the same direction as the fuel pipe of the fuel valve body.

FUEL STRAINER CLEANING

Drain the fuel from the fuel tank.
Loosen the fuel valve lock nut and remove the fuel valve.





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4. Fuel System

Remove the fuel strainer and O-ring.
Clean the fuel strainer.
Install the fuel strainer and O-ring onto the fuel valve.
Install the fuel valve and tighten the lock nut.

NOTE:

Do not over tighten lock nut

Fill the tank with gasoline and make sure there are no fuel leaks.

INSTALLATION

Install the fuel tank in the reverse order of removal.
Check the vent hole of the filler cap for blockage.
Make sure that there are no fuel leaks.

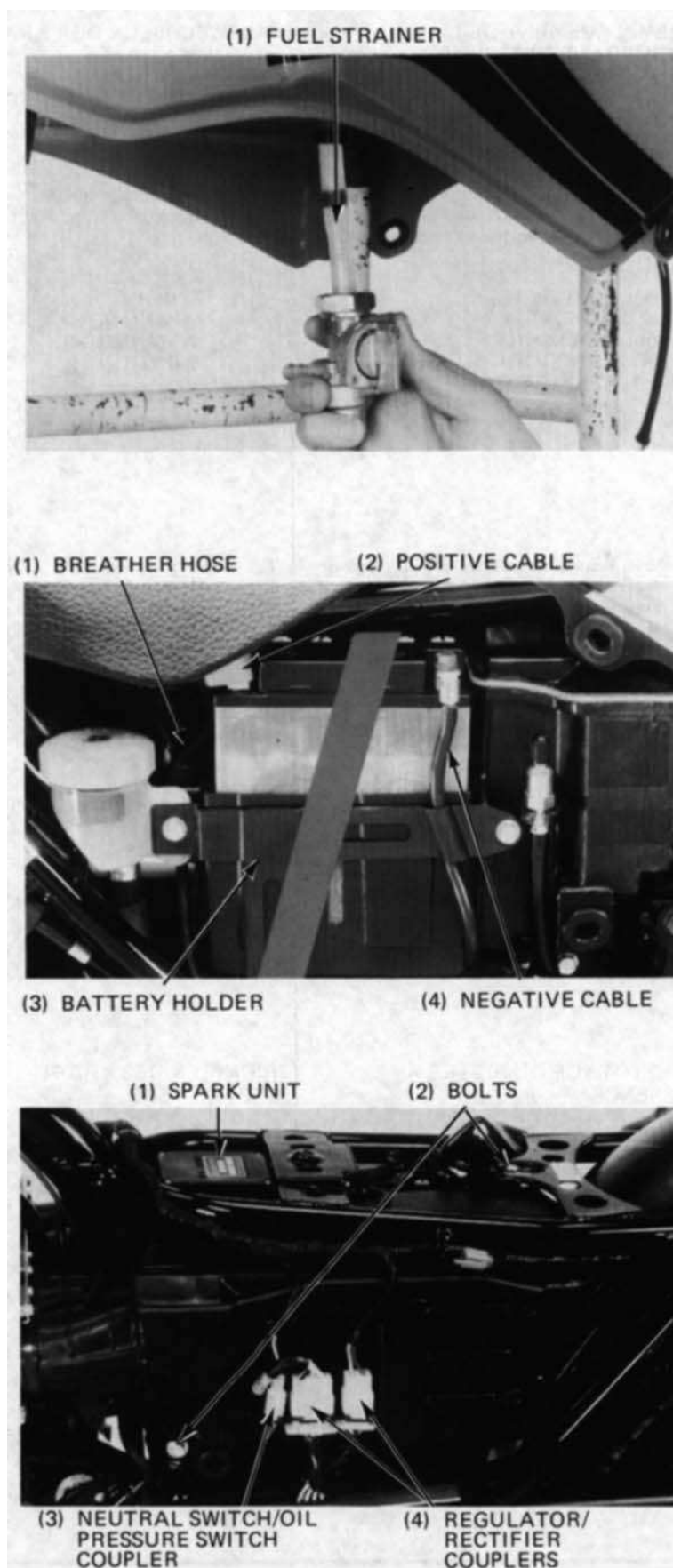
AIR CLEANER

Air cleaner case removal

Remove fuel tank (page 4-14).

Remove the battery holder.
Pull out the battery and disconnect the negative cable, then disconnect the positive cable.
Disconnect the battery breather hose and remove the battery.

Disconnect the regulator/rectifier couplers and neutral switch/oil pressure switch coupler.
Remove the spark unit from the air cleaner case.
Remove the bolts attaching the air cleaner case to the frame.
Loosen the air cleaner connecting tube bands.



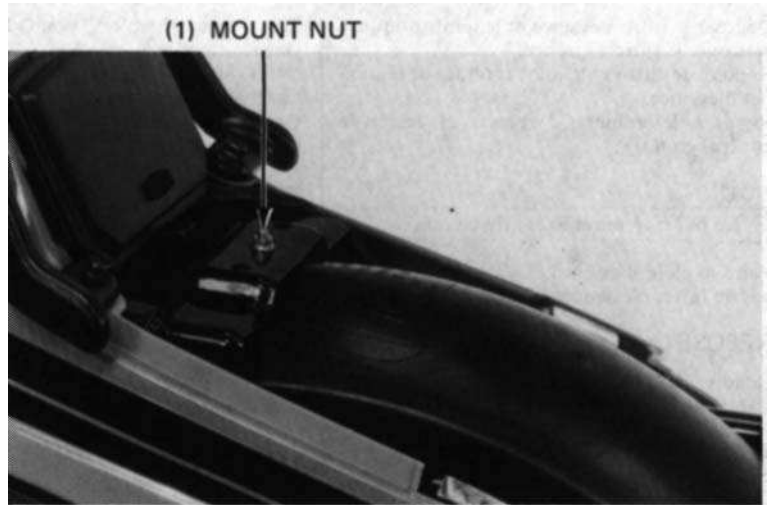


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4. Fuel System

Remove the rear wheel (page 14-3).
Unhook the rear fender tabs from the frame pipes and remove the mount nut and rear fender.
Remove the air cleaner case from the rear.

Check the air cleaner case seal rubbers for deterioration.



AIR CLEANER CASE INSTALLATION

Install the air cleaner case in the reverse order of the removal.

CRANKCASE VENTILATION SYSTEM

Check that the breather tube is not restricted.

